

Untitled

```
install.packages("seqinr")
install.packages("stringr")
```

```
upload_fasta <- function(fasta_filename) {
  library(seqinr)
  read.fasta(fasta_filename, seqtype = "AA", as.string = TRUE,
             set.attributes = FALSE)
}
```

```
upload_fasta("testfasta.txt")      # should return list of proteins A, B, C
```

```
$A
[1] "LVKLHHIIFESMLKDMQRRHRVW"
```

```
$B
[1] "ADEFQGSMQKIEACWQSYDVQF"
```

```
$C
[1] "MINEPFSWRLEFHLSEKRYDEIM"
```

```
upload_fasta("testfasta.txt")$A    # should return just protein A's sequence
```

```
[1] "LVKLHHIIFESMLKDMQRRHRVW"
```

```
trypsinize <- function(proteins) {
  library(stringr)
  lapply(proteins, str_split_1, pattern="(?<=R|K)")
}
```

```

proteins <- list(A="LVKLHHIIFESMLKDMQRRHRVW",
                B="ADEFQGSMQKIEACWQSYDVQF",
                C="MINEPFSWRLEFHLSEKRYDEIM")
trypsinize(proteins)

```

```

$A
[1] "LVK"          "LHHIIFESMLK" "DMQR"          "R"          "HR"
[6] "VW"

```

```

$B
[1] "ADEFQGSMQK"   "IEACWQSYDVQF"

```

```

$C
[1] "MINEPFSWR" "LEFHLSEK"   "K"          "YDEIM"

```

```

trypsinize(proteins)$A

```

```

[1] "LVK"          "LHHIIFESMLK" "DMQR"          "R"          "HR"
[6] "VW"

```

```

split_peptides <- function(peptides) {
  library(stringr)
  lapply(peptides, str_split, pattern="")
}

```

```

peptides <- list(A=c("LVK", "LHHIIFESMLK", "DMQR", "R", "HR", "VW"),
                B=c("ADEFQGSMQK", "IEACWQSYDVQF"),
                C=c("MINEPFSWR", "LEFHLSEK", "K", "YDEIM"))
split_peptides(peptides)

```

```

$A
$A[[1]]
[1] "L" "V" "K"

$A[[2]]
[1] "L" "H" "H" "I" "I" "F" "E" "S" "M" "L" "K"

$A[[3]]
[1] "D" "M" "Q" "R"

```

```
$A[[4]]  
[1] "R"
```

```
$A[[5]]  
[1] "H" "R"
```

```
$A[[6]]  
[1] "V" "W"
```

```
$B  
$B[[1]]  
[1] "A" "D" "E" "F" "Q" "G" "S" "M" "Q" "K"
```

```
$B[[2]]  
[1] "I" "E" "A" "C" "W" "Q" "S" "Y" "D" "V" "Q" "F"
```

```
$C  
$C[[1]]  
[1] "M" "I" "N" "E" "P" "F" "S" "W" "R"
```

```
$C[[2]]  
[1] "L" "E" "F" "H" "L" "S" "E" "R"
```

```
$C[[3]]  
[1] "K"
```

```
$C[[4]]  
[1] "Y" "D" "E" "I" "M"
```

```
split_peptides(peptides)$B
```

```
[[1]]  
[1] "A" "D" "E" "F" "Q" "G" "S" "M" "Q" "K"
```

```
[[2]]  
[1] "I" "E" "A" "C" "W" "Q" "S" "Y" "D" "V" "Q" "F"
```

```
splitpeptides_to_masses <- function(aa) {  
  aa_masses <- c(A=71.037, R=156.101, N=114.042, D=115.026, C=103.009,
```

```

      Q=128.058, E=129.042, G=57.021, H=137.058, I=113.084,
      L=113.084, K=128.094, M=131.040, F=147.068, P=97.052,
      S=87.032, T=101.047, W=186.079, Y=163.063, V=99.068)

peptide_masses <- aa
for(i in 1:length(aa)) {
  peptide_masses[[i]] <- lapply(aa[[i]], function(x) sum(aa_masses[x]))
}
lapply(peptide_masses, unlist)
}

```

```

aa <- list(A=list(c("L","V","K"),
                  c("L","H","H","I","I","F","E","S","M","L","K"),
                  c("D","M","Q","R")),
          B=list(c("A","D","E","F","Q","G","S","M","Q","K"),
                  c("I","E","A","C","W","Q","S","Y","D","V","Q","F")),
          C=list(c("M","I","N","E","P","F","S","W","R"),
                  c("L","E","F","H","L","S","E","R"),
                  c("Y","D","E","I","M")))
splitpeptides_to_masses(aa)

```

```

$A
[1] 340.246 1348.728 530.225

```

```

$B
[1] 1121.476 1469.624

```

```

$C
[1] 1160.540 1011.511 651.255

```

```

splitpeptides_to_masses(aa)$A # expect: 340.246 1348.728 530.225

```

```

[1] 340.246 1348.728 530.225

```

```

count_matching_masses <- function(protein_masses, sample) {
  df <- as.data.frame(sapply(protein_masses, function(x)
    sum(as.character(sample) %in% as.character(x))))
  names(df) <- "peptide_counts"
  return(df)
}

```

```

masses_list <- list(A=c(340.246, 1348.728, 530.225),
                   B=c(1121.476, 1469.624),
                   C=c(1160.540, 1011.511, 651.255))
sample <- c(340.246, 530.225, 1348.728)
count_matching_masses(masses_list, sample)

```

```

      peptide_counts
A                3
B                0
C                0

```

```

# expect: A=3, B=0, C=0

```

```

# Then test with a new random sample:
sample2 <- c(1121.476, 1160.540, 340.246)
count_matching_masses(masses_list, sample2)

```

```

      peptide_counts
A                1
B                1
C                1

```

```

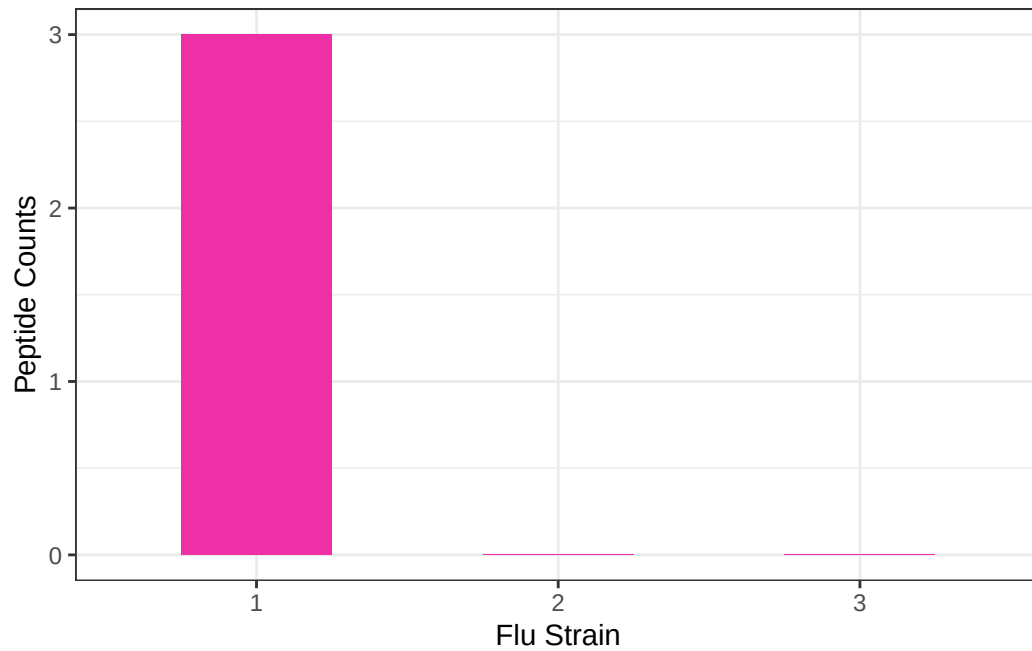
ggbarplot <- function(peptide_counts_table) {
  library(ggplot2)
  ggplot(peptide_counts_table) +
    aes(rownames(peptide_counts_table), peptide_counts) +
    geom_col(fill="maroon2", width=0.5) +
    theme_bw() +
    labs(x="Flu Strain", y="Peptide Counts")
}

```

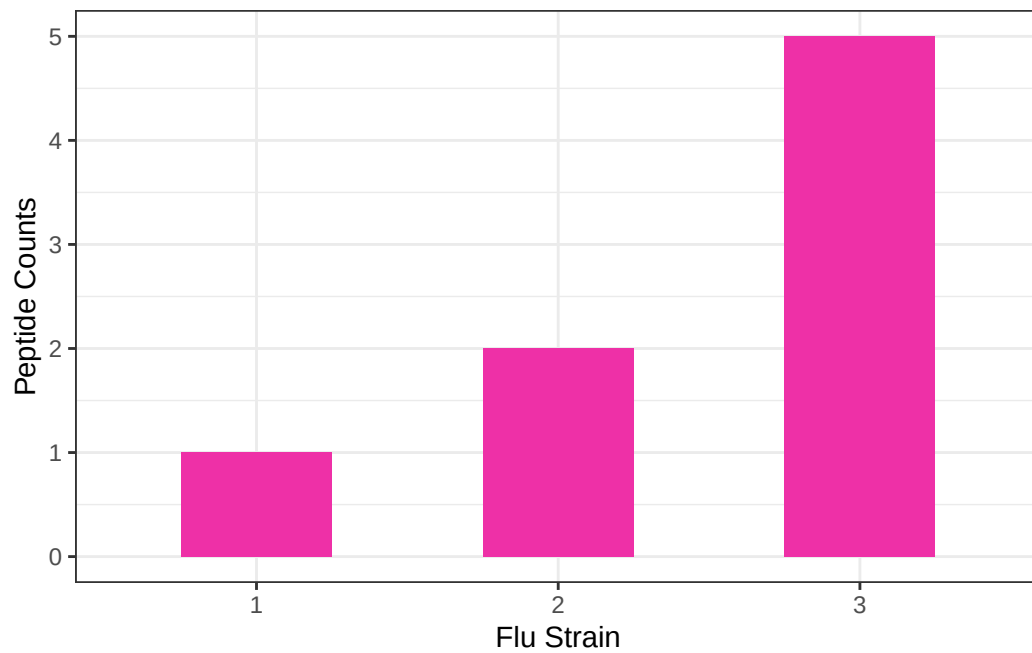
```

counts_df <- data.frame(peptide_counts=c(3, 0, 0), rownames=c("A", "B", "C"))
ggbarplot(counts_df)

```



```
# Also test with different numbers:  
counts_df2 <- data.frame(peptide_counts=c(1, 2, 5), rownames=c("A", "B", "C"))  
ggbarplot(counts_df2)
```



```
list.files(".")          # verify flufinder.R is there
```

```
[1] "finder.qmd"          "Flu Project.Rproj"
[3] "flu_proteins_database.txt" "flufinder.R"
[5] "lab19_Flu_files"      "lab19_Flu.html"
[7] "lab19_Flu.qmd"        "lab19_Flu.rmarkdown"
[9] "patient_sample.txt"    "README.md"
[11] "testfasta.txt"
```

```
source("flufinder.R")   # loads all 6 functions
```

```
# Load patient sample
patient_sample <- scan("patient_sample.txt")
```

```
# Step 1-2: upload and trypsinize
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize()
```

\$H5N1

```
[1] "DQICIGYHANNSTEQVDTIMEK"      "NVTVTHAQDILEK"
[3] "THNGK"                       "LCDLNGVK"
[5] "PLILR"                       "DCSVAGWLLGNPMCDEFINVPEWSYIVEK"
[7] "ASPANDLCYPGDFNDYEELK"        "HLLSR"
[9] "TNHFEK"                      "IQIIPK"
[11] "SSWSNHDASSGVSSACPYHGR"       "SSFFR"
[13] "NVVWLIK"                     "K"
[15] "NSAYPTIK"                    "R"
[17] "SYNNTNQEDLLVLWGIHHPNDAAEQTK" "LYQNPTTYISVGTSTLNQR"
[19] "LVPEIATR"                     "PK"
[21] "VNGQSGR"                     "MEFFWTILK"
[23] "PNDAINFESNGNFIAPEYAYK"      "IVK"
[25] "K"                            "GDSAIMK"
[27] "SELEYGNCNTK"                 "CQTPMGAINSSMPFHNIHPLTIGECPK"
[29] "YVK"                         "SNR"
[31] "LVLATGLR"                     "NTPQR"
[33] "ER"                           "R"
[35] "R"                            "K"
[37] "K"                            "R"
[39] ""
```

\$H3N2

```
[1] "QK"
```

[2] "LPGNDNSTATLCLGHHAVPNGTIVK"
[3] "TITNDQIEVTNATELVQSSSTGGICDSPHQILDGENCTLIDALLGDPQCDGFQNK"
[4] "K"
[5] "WDLFVER"
[6] "SK"
[7] "AYSNCYPYDVPDYASLR"
[8] "SLVASSGTLEFNNEFNFNWTGVTQNGTSSACK"
[9] "R"
[10] "R"
[11] "SNNSSFFSR"
[12] "LNWLTHLK"
[13] "FK"
[14] "YPALNVTMPNNEK"
[15] "FDK"
[16] "LYIWGVHHPGTDNDQISLYAQASGR"
[17] "ITVSTK"
[18] "R"
[19] "SQQTVIPSIGSR"
[20] "PR"
[21] "IR"
[22] "DVPSR"
[23] "ISIWWTIVK"
[24] "PGDILLINSTGNLIAPR"
[25] "GYFK"
[26] "IR"
[27] "SGK"
[28] "SSIMR"
[29] "SDAPIGK"
[30] "CNSECITPNGSIPNDK"
[31] "PFQNVNR"
[32] "ITYGACPR"
[33] "YVK"
[34] "QNTLK"
[35] "LATGMR"
[36] "NVPEK"
[37] "QTR"
[38] ""

\$H7N9

[1] "DK"
[2] "ICLGHHAVSNGTK"
[3] "VNTLTER"
[4] "GVEVVNATETVER"

[5] "TNIPR"
 [6] "ICSK"
 [7] "GK"
 [8] "R"
 [9] "TVDLGQCGLLGTTITGPPQCDQFLEFSADLIIR"
 [10] "R"
 [11] "EGSDVCYPGK"
 [12] "FVNEEALR"
 [13] "QILR"
 [14] "ESGGIDK"
 [15] "EAMGFTYSGIR"
 [16] "TNGATSACR"
 [17] "R"
 [18] "SGSSFYAEMK"
 [19] "WLLSNTDNAAFPQMTK"
 [20] "SYK"
 [21] "NTR"
 [22] "K"
 [23] "SPALIVWGIHHSVSTAEQTK"
 [24] "LYGSGNK"
 [25] "LVTVGSSNYQQSFVPSPGAR"
 [26] "PQVNGLSGR"
 [27] "IDFHWMLNPNNDTVTFSFNGAFIAPDR"
 [28] "ASFLR"
 [29] "GK"
 [30] "SMGIQSGVQVDANCEGDCYHSGGTIISNLPFQNIDSR"
 [31] "AVGK"
 [32] "CPR"
 [33] "YVK"
 [34] "QR"
 [35] "SLLLATGMK"
 [36] "NVPEIPK"
 [37] "GR"
 [38] ""

\$H2N2

[1] "DQICIGYHANNSTEK"	"VDTILER"
[3] "NVTVTHAK"	"DILEK"
[5] "THNGK"	"LCK"
[7] "LNGIPPLELGDCSIAGWLLGNPECDR"	"LLSVPEWSYIMEK"
[9] "ENPR"	"YSLCYPGSFNDYEELK"
[11] "HLLSSVK"	"HFEK"
[13] "VK"	"ILPK"

[15]	"DR"	"WTQHTTTGGSWACAVSGK"
[17]	"PSFFR"	"NMVWLTR"
[19]	"K"	"GSNYPVAK"
[21]	"GSYNNTSGEQMLIIWGVHHPNDEAEQR"	"ALYQNVGTIVSVATSTLYK"
[23]	"R"	"SIPEIAAR"
[25]	"PK"	"VNLGR"
[27]	"R"	"MEFSWTLLDMWDTINFESTGNLVAPEYGFK"
[29]	"ISK"	"R"
[31]	"GSSGIMK"	"TEGTLENCETK"
[33]	"CQTPLGAINTTLPFHNHPLTIGECPK"	"YVK"
[35]	"SEK"	"LVLATGLR"
[37]	"NVPQIESR"	" "

\$H1N1

[1]	"DTICIGYHANNSTDTVDTVLEK"	"NVTVTHSVNLLED SHNGK"
[3]	"LCR"	"LK"
[5]	"GIAPLQLGK"	"CNIAGWLLGNPECDPLLPVR"
[7]	"SWSYIVETPNSENGICYPGDFIDYEELR"	"EQLSSVSSFER"
[9]	"FEIFPK"	"ESSWPNHNNTTK"
[11]	"GVTAACSHAGK"	"SSFYR"
[13]	"NLLWLTEK"	"EGSYPK"
[15]	"LK"	"NSYVNK"
[17]	"K"	"GK"
[19]	"EVLVLWGIHHPNSK"	"DQQNIYQENAYVSVVTSNYNR"
[21]	"R"	"FTPEIAER"
[23]	"PK"	"VR"
[25]	"DQAGR"	"MNYWTLK"
[27]	"PGDTIIFEANGNLIAPR"	"YAFALSR"
[29]	"GFGSGIITSNASMHECNTK"	"CQTPLGAINSSLPFQNIHPVTIGECPK"
[31]	"YVR"	"SAK"
[33]	"LR"	"MVTGLR"
[35]	"NIPSIQSR"	" "

\$`B/Lee`

[1]	"MK"
[2]	"AIIVLLMVVTSNADR"
[3]	"ICTGITSSNSPHVVK"
[4]	"TATQGEVNVTVIPLTTTPTK"
[5]	"SHFANLK"
[6]	"GTQTR"
[7]	"GK"
[8]	"LCPNCFNCTDL DVALGR"
[9]	"PK"

[10] "CMGNTPSAK"
[11] "VSILHEVK"
[12] "PATSGCFPIMHDR"
[13] "TK"
[14] "IR"
[15] "QLPNLLR"
[16] "GYENIR"
[17] "LSTSNVINTETAPGGPYK"
[18] "VGTSGSCPNVANGNGFFNTMAWVIPK"
[19] "DNNK"
[20] "TAINPVTVEVPYICSEGEDQITVWGFHSDDK"
[21] "TQMER"
[22] "LYGDSNPQK"
[23] "FTSSANGVTTHYVSQIGGFNPQTEDEGLK"
[24] "QSGR"
[25] "IVVDYMVQK"
[26] "PGK"
[27] "TGTIVYQR"
[28] "GILLPQK"
[29] "VWCASGR"
[30] "SK"
[31] "VIK"
[32] "GSLPLIGEADCLHEK"
[33] "YGGLNK"
[34] "SK"
[35] "PYYTGEHAK"
[36] "AIGNCPIWVK"
[37] "TPLK"
[38] "LANGTK"
[39] "YR"
[40] "PPAK"
[41] "LLK"
[42] "ER"
[43] "GFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLK"
[44] "STQEAINK"
[45] "ITK"
[46] "NLNYLSELEVK"
[47] "NLQR"
[48] "LSGAMNELHDEILELDEK"
[49] "VDDLRL"
[50] "ADTISSQIELAVLLSNEGIINSEDEHLLALER"
[51] "K"
[52] "LK"

[53] "K"
[54] "MLGPSAVEIGNGCFETK"
[55] "HK"
[56] "CNQTCLDR"
[57] "IAAGTFNAGDFSLPTFDSL NITAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFIVYMVSR"
[58] "DNVSCSICL"

\$H9N2

[1] "METISLITILLVVTASNADK"
[2] "ICIGHQSTNSTETVDTLTETNVPVTHAK"
[3] "ELLHTEHNGMLCATSLGHPLILDCTIEGLVYGNPSCDLLLGGR"
[4] "EWSYIVER"
[5] "SSAVNGTCYPGNVENLEELR"
[6] "TLFSSASSYQR"
[7] "IQIFPDTTWNVTYTGTSR"
[8] "ACSGSFYR"
[9] "SMR"
[10] "WLTQK"
[11] "SGFYVPQDAQYTNNR"
[12] "GK"
[13] "SILFVWGIHHPPTYTEQTNLYIR"
[14] "NDTTTSVTTEDLNR"
[15] "TFK"
[16] "PVIGPR"
[17] "PLVNGLQGR"
[18] "IDYYWSVLK"
[19] "PGQTLR"
[20] "VR"
[21] "SNGNLIAPWYGHVLSGGSHGR"
[22] "ILK"
[23] "TDLK"
[24] "GGNCVVQCQTEK"
[25] "GGLNSTLPFHNISK"
[26] "YAFGTCPK"
[27] "YVR"
[28] "VNSLK"
[29] "LAVGLR"
[30] "NVPAR"
[31] "SSR"
[32] "GLFGAIAGFIEGGWPGLVAGWYGFQHSNDQGVGMAADR"
[33] "DSTQK"
[34] "AIDK"
[35] "ITSK"

```

[36] "VNNIVDK"
[37] "MNK"
[38] "QYEIDHEFSEVETR"
[39] "LNMINNK"
[40] "IDDDIQDVWAYNAELLVLENQK"
[41] "TLDEHDANVNNLYNK"
[42] "VK"
[43] "R"
[44] "ALGSNAMEDGK"
[45] "GCFELYHK"
[46] "CDDQCMETIR"
[47] "NGTYNR"
[48] "R"
[49] "K"
[50] "YR"
[51] "EESR"
[52] "LER"
[53] "QK"
[54] "IEGVK"
[55] "LESEGTYSK"
[56] "ILTIYSTVASSLVLAMGFAAFLWAMSNGSCR"
[57] "CNICI"

```

```

# Step 1-4: add splitting and mass calculation
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize() |>
  split_peptides() |> splitpeptides_to_masses()

```

\$H5N1

```

[1] 2490.104 1448.755 537.262 842.428 592.405 3295.490 2241.942 606.359
[9] 756.351 692.456 2172.896 624.301 852.519 128.094 874.451 156.101
[17] 3088.449 2137.072 879.515 225.146 698.343 1195.606 2355.070 340.246
[25] 128.094 702.334 1238.518 2904.341 390.225 357.175 823.526 596.300
[33] 285.143 156.101 156.101 128.094 128.094 156.101 0.000

```

\$H3N2

```

[1] 256.152 2510.257 5799.656 128.094 945.468 215.126 1977.850 3217.449
[9] 156.101 156.101 939.417 1005.572 275.162 1471.705 390.188 2779.336
[17] 629.372 156.101 1253.669 253.153 269.185 554.279 1103.635 1744.977
[25] 495.246 269.185 272.147 574.289 668.346 1672.710 855.431 861.414
[33] 390.225 584.325 629.330 567.298 385.206 0.000

```

\$H7N9

[1]	243.120	1317.655	813.431	1383.693	581.326	431.219	185.115	156.101
[9]	3530.729	156.101	1035.428	958.484	510.327	686.320	1212.556	861.373
[17]	156.101	1087.461	1817.869	378.189	371.190	128.094	2142.114	719.357
[25]	2075.036	908.479	3119.483	574.322	185.115	3893.725	355.220	356.162
[33]	390.225	284.159	914.523	777.434	213.122	0.000		

\$H2N2

[1]	1673.739	826.452	850.461	598.330	537.262	344.187	2733.315	1575.796
[9]	496.237	1908.817	764.452	541.262	227.162	451.314	271.127	1858.834
[17]	634.321	900.461	128.094	816.409	3063.376	2059.055	156.101	837.469
[25]	225.146	596.337	156.101	3522.601	328.210	156.101	660.324	1205.517
[33]	2882.443	390.225	344.168	823.526	923.479	0.000		

\$H1N1

[1]	2390.094	1944.955	372.194	241.178	877.535	2161.073	3277.443	1249.591
[9]	761.408	1281.567	982.461	640.296	997.556	661.304	241.178	705.341
[17]	128.094	185.115	1696.898	2600.178	156.101	943.473	225.146	255.169
[25]	527.243	1212.596	1778.924	808.422	1934.853	2845.413	418.232	286.163
[33]	269.185	657.361	895.485	0.000				

\$`B/Lee`

[1]	259.134	1595.902	1523.770	2109.120	797.415	543.274	185.115	1834.809
[9]	225.146	889.374	905.530	1412.627	229.141	269.185	834.505	732.353
[17]	1829.907	2649.234	471.204	3430.587	645.288	1002.472	3065.422	428.212
[25]	1075.569	282.167	918.489	749.477	759.347	215.126	340.246	1562.770
[33]	632.325	215.126	1046.477	1081.570	439.277	584.325	319.164	393.235
[41]	354.262	285.143	3925.863	871.436	342.225	1302.677	511.285	2036.964
[49]	598.305	3473.778	128.094	241.178	128.094	1733.804	265.152	933.376
[57]	6667.332	934.386						

\$H9N2

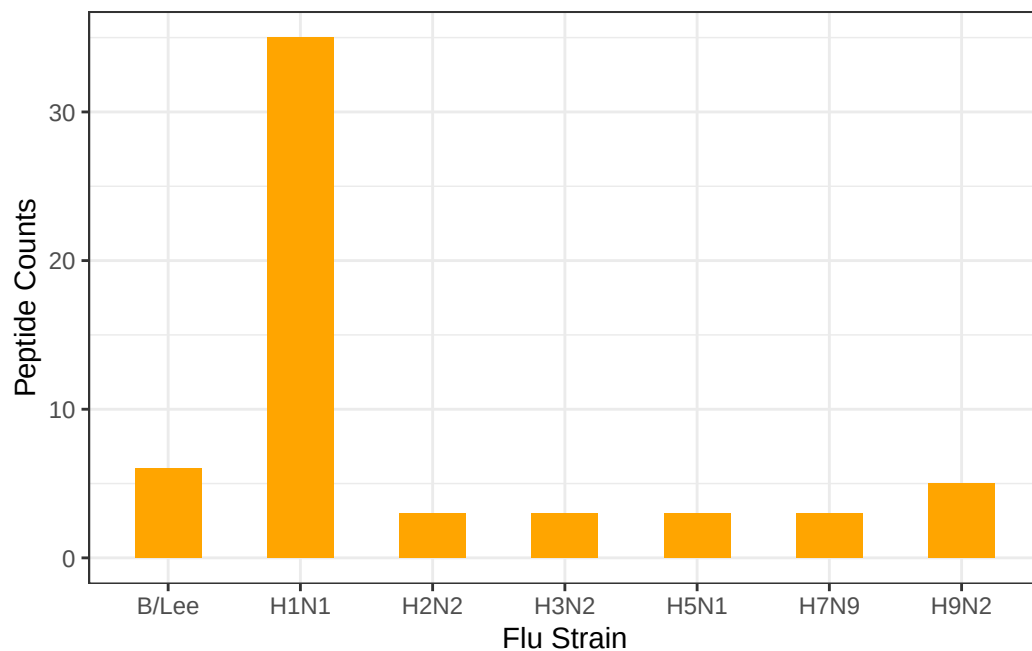
[1]	2113.163	2977.430	4657.262	1062.511	2132.971	1227.586	2081.013	871.363
[9]	374.173	656.362	1740.778	185.115	2766.418	1547.698	376.209	619.378
[17]	934.531	1167.593	652.363	255.169	2160.053	354.262	457.251	1246.537
[25]	1465.761	867.391	418.232	541.320	609.395	537.300	330.165	3889.827
[33]	559.257	427.241	429.257	782.424	373.176	1875.857	827.428	2711.370
[41]	1740.797	227.162	156.101	1073.476	977.439	1194.442	705.316	156.101
[49]	128.094	319.164	501.217	398.227	256.152	526.309	907.425	3408.699
[57]	546.228							

```
# Step 1-5: count matches against patient sample
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize() |>
```

```
split_peptides() |> splitpeptides_to_masses() |>
count_matching_masses(sample=patient_sample)
```

	peptide_counts
H5N1	3
H3N2	3
H7N9	3
H2N2	3
H1N1	35
B/Lee	6
H9N2	5

```
# Full pipeline with plot
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize() |>
split_peptides() |> splitpeptides_to_masses() |>
count_matching_masses(sample=patient_sample) |> ggbarplot()
```



```
list.files("/Users/abyperez/Library/Mobile Documents/com~apple~CloudDocs/BIMM 143/Flu Project")
```

```
[1] "finder.qmd" "Flu Project.Rproj"
[3] "flu_proteins_database.txt" "flufinder.R"
```

```
[5] "lab19_Flu_files"          "lab19_Flu.html"
[7] "lab19_Flu.qmd"           "lab19_Flu.rmarkdown"
[9] "patient_sample.txt"       "README.md"
[11] "testfasta.txt"
```

```
source("/Users/abyperez/Library/Mobile Documents/com~apple~CloudDocs/BIMM 143/Flu Project/f
```

```
# Using scan() to generate a vector from a text file with space separated numbers
patient_sample <- scan("patient_sample.txt")
patient_sample
```

```
[1] 2390.094 1944.955 372.194 241.178 877.535 2161.073 3277.443 1249.591
[9] 761.408 1281.567 982.461 640.296 997.556 661.304 241.178 705.341
[17] 128.094 185.115 1696.898 2600.178 156.101 943.473 225.146 255.169
[25] 527.243 1212.596 1778.924 808.422 1934.853 2845.413 418.232 286.163
[33] 269.185 657.361 895.485
```

```
# Inspecting the flu protein database
upload_fasta("flu_proteins_database.txt")
```

\$H5N1

```
[1] "DQICIGYHANNSTEQVDTIMEKNVTVTHAQDILEKTHNGKLCDLNGVKPLILRDCSVAGWLLGNPMCDEFINVPEWSYIVEKASPAN"
```

\$H3N2

```
[1] "QKLPGNDNSTATLCLGHHAVPNGTIVKTITNDQIEVTNATELVQSSSTGGICDSPHQILDGENCTLIDALLGDPQCDGFGNKKWDLF"
```

\$H7N9

```
[1] "DKICLGHHAVSNGTKVNTLTERGVEVVNATETVERTNIPRICSKGKRTVDLGQCGLLGTTITGPPQCDQFLEFSADLIIERREGSDVC"
```

\$H2N2

```
[1] "DQICIGYHANNSTKVDITLERNVTVTHAKDILEKTHNGKLCCKLNGIPPELGDCSIAGWLLGNPECDRLLSVPEWSYIMEKENPRY"
```

\$H1N1

```
[1] "DTICIGYHANNSTDTVDTVLEKNVTVTHSVNLLED SHNGKLCRLKGIAPLQLGKCNIAGWLLGNPECDPLLPVRSWSYIVETPSEN"
```

\$`B/Lee`

```
[1] "MKAIIVLLMVVTSNADRICTGITSSNSPHVVKATATQGEVNVTVGIPLTTTPTKSHFANLKGTQTRGKLCPCNCFNCTDL DVALGRPKC"
```

\$H9N2

```
[1] "METISLITILLVVTASNADKICIGHQSTNSTETVDTLTETNVPVTHAKELLHTEHNGMLCATSLGHPLILDCTIEGLVYGNPSCDL"
```

```
# Piping the flu protein database to trypsinize()
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize()
```

\$H5N1

[1] "DQICIGYHANNSTEQVDTIMEK"	"NVTVTHAQDILEK"
[3] "THNGK"	"LCDLNGVK"
[5] "PLILR"	"DCSVAGWLLGNPMCDEFINPEWSYIVEK"
[7] "ASPANDLCYPGDFNDYEELK"	"HLLSR"
[9] "TNHFKEK"	"IQIIPK"
[11] "SSWSNHDASSGVSSACPYHGR"	"SSFFR"
[13] "NVVWLIK"	"K"
[15] "NSAYPTIK"	"R"
[17] "SYNNTNQEDLLVLWGIHPNDAAEQTK"	"LYQNPTTYISVGTSTLNQR"
[19] "LVPEIATR"	"PK"
[21] "VNGQSGR"	"MEFFWTILK"
[23] "PNDAINFESNGNFIAPEYAYK"	"IVK"
[25] "K"	"GDSAIMK"
[27] "SELEYGNCNTK"	"CQTPMGAINSSMPFHNIHPLTIGECPK"
[29] "YVK"	"SNR"
[31] "LVLATGLR"	"NTPQR"
[33] "ER"	"R"
[35] "R"	"K"
[37] "K"	"R"
[39] ""	

\$H3N2

[1] "QK"
[2] "LPGNDNSTATLCLGHHAVPNGTIVK"
[3] "TITNDQIEVTNATELVQSSSTGGICDSPHQILDGENCTLIDALLGDPQCDGFQNK"
[4] "K"
[5] "WDLFVER"
[6] "SK"
[7] "AYSNCYPYDVPDYASLR"
[8] "SLVASSGTLEFNNEFNTGTQNGTSSACK"
[9] "R"
[10] "R"
[11] "SNNSFFSR"
[12] "LNWLTHLK"
[13] "FK"
[14] "YPALNVTMPNNEK"
[15] "FDK"
[16] "LYIWGVHHPGTDNDQISLYAQASGR"

[17] "ITVSTK"
[18] "R"
[19] "SQQTVIPSIGSR"
[20] "PR"
[21] "IR"
[22] "DVPSR"
[23] "ISYWTIVK"
[24] "PGDILLINSTGNLIAPR"
[25] "GYFK"
[26] "IR"
[27] "SGK"
[28] "SSIMR"
[29] "SDAPIGK"
[30] "CNSECITPNGSIPNDK"
[31] "PFQNVNR"
[32] "ITYGACPR"
[33] "YVK"
[34] "QNTLK"
[35] "LATGMR"
[36] "NVPEK"
[37] "QTR"
[38] ""

\$H7N9

[1] "DK"
[2] "ICLGHHAVSNGTK"
[3] "VNTLTER"
[4] "GVEVVNATETVER"
[5] "TNIPR"
[6] "ICSK"
[7] "GK"
[8] "R"
[9] "TVDLGQCGLLGTITGPPQCDQFLEFSADLIIR"
[10] "R"
[11] "EGSDVCYPGK"
[12] "FVNEEALR"
[13] "QILR"
[14] "ESGGIDK"
[15] "EAMGFTYSGIR"
[16] "TNGATSACR"
[17] "R"
[18] "SGSSFYAEMK"
[19] "WLLSNTDNAAFPMQTK"

[20] "SYK"
 [21] "NTR"
 [22] "K"
 [23] "SPALIVWGIHHSVSTAEQTK"
 [24] "LYGSGNK"
 [25] "LVTVGSSNYQQSFVPSPGAR"
 [26] "PQVNGLSGR"
 [27] "IDFHWLMLNPNDTVTFSFNAGFIAPDR"
 [28] "ASFLR"
 [29] "GK"
 [30] "SMGIQSGVQVDANCEGDCYHSGGTIISNLPFQNIDSR"
 [31] "AVGK"
 [32] "CPR"
 [33] "YVK"
 [34] "QR"
 [35] "SLLLATGMK"
 [36] "NVPEIPK"
 [37] "GR"
 [38] ""

\$H2N2

[1] "DQICIGYHANNSTEK"	"VDTILER"
[3] "NVTVTHAK"	"DILEK"
[5] "THNGK"	"LCK"
[7] "LNGIPPLELGDCSIAGWLLGNPECDR"	"LLSVPEWSYIMEK"
[9] "ENPR"	"YSLCYPGSFNDYEELK"
[11] "HLLSSVK"	"HF EK"
[13] "VK"	"ILPK"
[15] "DR"	"WTQHTTTGGSWACAVSGK"
[17] "PSFFR"	"NMVWLTR"
[19] "K"	"GSNYPVAK"
[21] "GSYNNTSGEQMLIIWGVHHPNDEAEQR"	"ALYQNVGTYVSVATSTLYK"
[23] "R"	"SIPEIAAR"
[25] "PK"	"VNGLGR"
[27] "R"	"MEFSWTLLDMWDTINFESTGNLVAPEYGFK"
[29] "ISK"	"R"
[31] "GSSGIMK"	"TEGTLENCETK"
[33] "CQTPLGAINTTLPFHNVHPLTIGECPK"	"YVK"
[35] "SEK"	"LVLATGLR"
[37] "NVPQIESR"	" "

\$H1N1

[1] "DTICIGYHANNSTDVDTVLEK"	"NVTVTHSVNLLLED SHNGK"
-----------------------------	------------------------

[3]	"LCR"	"LK"
[5]	"GIAPLQLGK"	"CNIAGWLLGNPECDPLLPVR"
[7]	"SWSYIVETPNSENGICYPGDFIDYEELR"	"EQLSSVSSFER"
[9]	"FEIFPK"	"ESSWPNHNTTK"
[11]	"GVTAACSHAGK"	"SSFYR"
[13]	"NLLWLTEK"	"EGSYPK"
[15]	"LK"	"NSYVNK"
[17]	"K"	"GK"
[19]	"EVLVLWGIHHPNSK"	"DQQNIYQENAYVSVVTSNYNR"
[21]	"R"	"FTPEIAER"
[23]	"PK"	"VR"
[25]	"DQAGR"	"MNYWTLK"
[27]	"PGDTIIFEANGNLIAPR"	"YAFALSR"
[29]	"GFGSGIITSNASMHECNTK"	"CQTPLGAINSSLPFQNIHPVTIGECPK"
[31]	"YVR"	"SAK"
[33]	"LR"	"MVTGLR"
[35]	"NIPSIQSR"	" "

\$`B/Lee`

[1]	"MK"
[2]	"AIIVLLMVVTSNADR"
[3]	"ICTGITSSNSPHVVK"
[4]	"TATQGEVNVTVIPLTTTPTK"
[5]	"SHFANLK"
[6]	"GTQTR"
[7]	"GK"
[8]	"LCPNCFNCTDLVALGR"
[9]	"PK"
[10]	"CMGNTPSAK"
[11]	"VSILHEVK"
[12]	"PATSGCFPIHDR"
[13]	"TK"
[14]	"IR"
[15]	"QLPNLLR"
[16]	"GYENIR"
[17]	"LSTSNVINTETAPGGPYK"
[18]	"VGTSGSCPNAVANGFFNTMAWVIPK"
[19]	"DNNK"
[20]	"TAINPVTVEVPYICSEGEDQITVWGFHSDDK"
[21]	"TQMER"
[22]	"LYGDSNPQK"
[23]	"FTSSANGVTTHYVSQIGGFNPQTEDEGLK"
[24]	"QSGR"

[25] "IVVDYMVQK"
[26] "PGK"
[27] "TGTIVYQR"
[28] "GILLPQK"
[29] "VWCASGR"
[30] "SK"
[31] "VIK"
[32] "GSLPLIGEADCLHEK"
[33] "YGGLNK"
[34] "SK"
[35] "PYYTGEHAK"
[36] "AIGNCPIWVK"
[37] "TPLK"
[38] "LANGTK"
[39] "YR"
[40] "PPAK"
[41] "LLK"
[42] "ER"
[43] "GFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLK"
[44] "STQEAINK"
[45] "ITK"
[46] "NLNYLSELEVK"
[47] "NLQR"
[48] "LSGAMNELHDEILELDEK"
[49] "VDDLRL"
[50] "ADTISSQIELAVLLSNEGIINSEDEHLLALER"
[51] "K"
[52] "LK"
[53] "K"
[54] "MLGPSAVEIGNGCFETK"
[55] "HK"
[56] "CNQTCLDR"
[57] "IAAGTFNAGDFSLPTFDSLNTAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFIVYMVSR"
[58] "DNVSCSICL"

\$H9N2

[1] "METISLITILLVVTASNADK"
[2] "ICIGHQSTNSTETVDTLTETNVPVTHAK"
[3] "ELLHTEHNGMLCATSLGHPLILDCTIEGLVYGNPSCDLLLGGR"
[4] "EWSYIVER"
[5] "SSAVNGTCYPGNVENLEELR"
[6] "TLFSSASSYQR"
[7] "IQIFPDTTWNVTYTGTSR"

[8] "ACSGSFYR"
[9] "SMR"
[10] "WLTQK"
[11] "SGFYPVQDAQYTNNR"
[12] "GK"
[13] "SILFVWGIHHPPTYTEQTNLYIR"
[14] "NDTTTSVTTEDLNR"
[15] "TFK"
[16] "PVIGPR"
[17] "PLVNGLQGR"
[18] "IDYYWSVLK"
[19] "PGQTLR"
[20] "VR"
[21] "SNGNLIAPWYGHVLSGGSHGR"
[22] "ILK"
[23] "TDLK"
[24] "GGNCVVQCQTEK"
[25] "GGLNSTLPFHNISK"
[26] "YAFGTCPK"
[27] "YVR"
[28] "VNSLK"
[29] "LAVGLR"
[30] "NVPAR"
[31] "SSR"
[32] "GLFGAIAGFIEGGWPGLVAGWYGFQHSNDQGVGMAADR"
[33] "DSTQK"
[34] "AIDK"
[35] "ITSK"
[36] "VNNIVDK"
[37] "MNK"
[38] "QYEIIDHEFSEVETR"
[39] "LNMINNK"
[40] "IDDQIQDVWAYNAELLVLENQK"
[41] "TLDEHDANVNNLYNK"
[42] "VK"
[43] "R"
[44] "ALGSNAMEDGK"
[45] "GCFELYHK"
[46] "CDDQCMETIR"
[47] "NGTYNR"
[48] "R"
[49] "K"
[50] "YR"

```

[51] "EESR"
[52] "LER"
[53] "QK"
[54] "IEGVK"
[55] "LESEGTYSK"
[56] "ILTIYSTVASSLVLAMGFAAFLFWAMSNGSCR"
[57] "CNICI"

```

Q1: What do the output strings represent and why do they all end in R or K?

```

# Piping until splitpeptides_to_masses()
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize() |>
split_peptides() |> splitpeptides_to_masses()

```

\$H5N1

```

[1] 2490.104 1448.755 537.262 842.428 592.405 3295.490 2241.942 606.359
[9] 756.351 692.456 2172.896 624.301 852.519 128.094 874.451 156.101
[17] 3088.449 2137.072 879.515 225.146 698.343 1195.606 2355.070 340.246
[25] 128.094 702.334 1238.518 2904.341 390.225 357.175 823.526 596.300
[33] 285.143 156.101 156.101 128.094 128.094 156.101 0.000

```

\$H3N2

```

[1] 256.152 2510.257 5799.656 128.094 945.468 215.126 1977.850 3217.449
[9] 156.101 156.101 939.417 1005.572 275.162 1471.705 390.188 2779.336
[17] 629.372 156.101 1253.669 253.153 269.185 554.279 1103.635 1744.977
[25] 495.246 269.185 272.147 574.289 668.346 1672.710 855.431 861.414
[33] 390.225 584.325 629.330 567.298 385.206 0.000

```

\$H7N9

```

[1] 243.120 1317.655 813.431 1383.693 581.326 431.219 185.115 156.101
[9] 3530.729 156.101 1035.428 958.484 510.327 686.320 1212.556 861.373
[17] 156.101 1087.461 1817.869 378.189 371.190 128.094 2142.114 719.357
[25] 2075.036 908.479 3119.483 574.322 185.115 3893.725 355.220 356.162
[33] 390.225 284.159 914.523 777.434 213.122 0.000

```

\$H2N2

```

[1] 1673.739 826.452 850.461 598.330 537.262 344.187 2733.315 1575.796
[9] 496.237 1908.817 764.452 541.262 227.162 451.314 271.127 1858.834
[17] 634.321 900.461 128.094 816.409 3063.376 2059.055 156.101 837.469
[25] 225.146 596.337 156.101 3522.601 328.210 156.101 660.324 1205.517
[33] 2882.443 390.225 344.168 823.526 923.479 0.000

```

\$H1N1

```
[1] 2390.094 1944.955 372.194 241.178 877.535 2161.073 3277.443 1249.591
[9] 761.408 1281.567 982.461 640.296 997.556 661.304 241.178 705.341
[17] 128.094 185.115 1696.898 2600.178 156.101 943.473 225.146 255.169
[25] 527.243 1212.596 1778.924 808.422 1934.853 2845.413 418.232 286.163
[33] 269.185 657.361 895.485 0.000
```

\$`B/Lee`

```
[1] 259.134 1595.902 1523.770 2109.120 797.415 543.274 185.115 1834.809
[9] 225.146 889.374 905.530 1412.627 229.141 269.185 834.505 732.353
[17] 1829.907 2649.234 471.204 3430.587 645.288 1002.472 3065.422 428.212
[25] 1075.569 282.167 918.489 749.477 759.347 215.126 340.246 1562.770
[33] 632.325 215.126 1046.477 1081.570 439.277 584.325 319.164 393.235
[41] 354.262 285.143 3925.863 871.436 342.225 1302.677 511.285 2036.964
[49] 598.305 3473.778 128.094 241.178 128.094 1733.804 265.152 933.376
[57] 6667.332 934.386
```

\$H9N2

```
[1] 2113.163 2977.430 4657.262 1062.511 2132.971 1227.586 2081.013 871.363
[9] 374.173 656.362 1740.778 185.115 2766.418 1547.698 376.209 619.378
[17] 934.531 1167.593 652.363 255.169 2160.053 354.262 457.251 1246.537
[25] 1465.761 867.391 418.232 541.320 609.395 537.300 330.165 3889.827
[33] 559.257 427.241 429.257 782.424 373.176 1875.857 827.428 2711.370
[41] 1740.797 227.162 156.101 1073.476 977.439 1194.442 705.316 156.101
[49] 128.094 319.164 501.217 398.227 256.152 526.309 907.425 3408.699
[57] 546.228
```

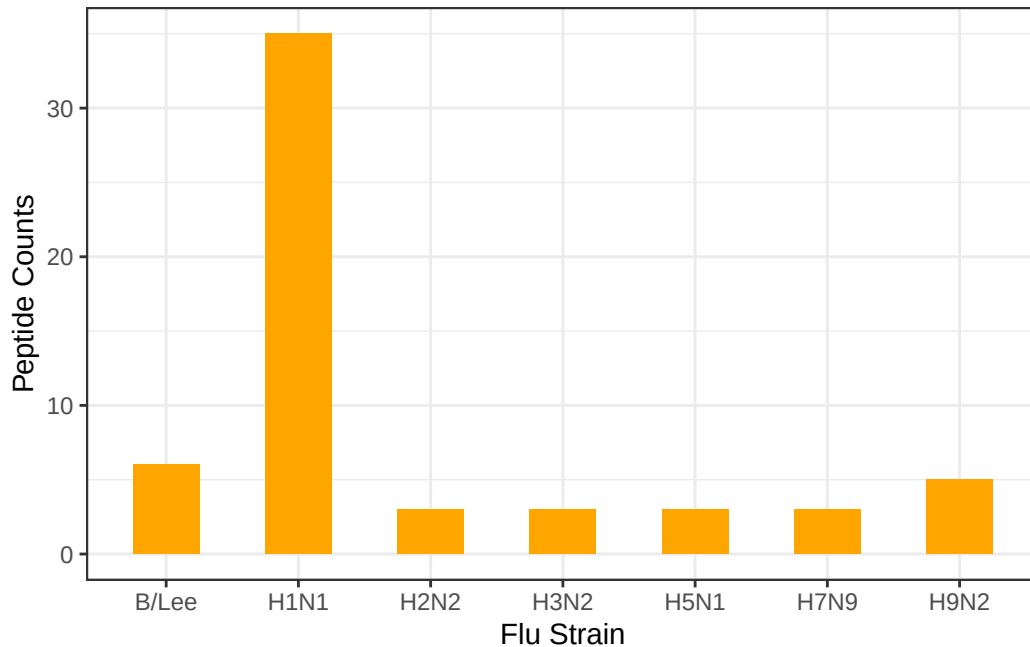
Q2: What do all the numbers represent?

```
# Piping until count_matching_masses()
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize() |>
split_peptides() |> splitpeptides_to_masses() |>
count_matching_masses(sample=patient_sample)
```

	peptide_counts
H5N1	3
H3N2	3
H7N9	3
H2N2	3
H1N1	35
B/Lee	6
H9N2	5

Q3: What do the numbers in the table represent and why does the `count_matching_masses()` function need the sample input in addition to the standard input from the pipeline?

```
# Using piping to consecutively run all functions of the flufinder script
"flu_proteins_database.txt" |> upload_fasta() |> trypsinize() |>
split_peptides() |> splitpeptides_to_masses() |>
count_matching_masses(sample=patient_sample) |> ggbarplot()
```



Q4: Which flu strain do you order an mRNA vaccine to be mass produced for?

Q5: Why do the other flu strains show a subset of matches?